Scenario 4D: Short time-series stress test

MSSTNB fixed-gamma posterior performance under T = 30

2026-06-20

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1 Purpose and main conclusion

Scenario 4D is a short time-series stress test for the dynamic MSSTNB model. The scenario keeps the main dynamic covariate, spatial, dispersion, and latent-risk structure used in the validation settings, but reduces the temporal horizon to $T = 30$. The goal is to isolate the effect of limited temporal replication on posterior recovery and MCMC stability.

The main finding is that the model remains stable and recovers the regression effects accurately under $T = 30$. All 10 fitted replicates are classified as stable. Regression recovery is strong: the average posterior means are close to the true values for β_0, β_1, and β_2, and the empirical 95% coverage is 1 for all three coefficients. The primary effect of the short time series is weaker recovery of latent-path shape by correlation-based summaries, especially for increments in $\log \lambda$, while scale-based recovery measured by RMSE remains good.

2 Scenario design

2.1 What is changed in Scenario 4D?

The reference dynamic validation scenarios use $T = 100$. Scenario 4D reduces the temporal horizon to $T = 30$, so each replicate has only 270 fitted level-1 time-region cells when $n_1 = 9$. This directly reduces temporal replication for learning the latent dynamic paths and temporal covariate effects.

The data-generating mechanism otherwise keeps the main dynamic structure. The covariates are standardized time-varying covariates. In the fitted simulation data, x_2 is generated using a continuous-time AR process, and the summary diagnostics below confirm that x_2 varies over time within each region.

2.2 Target question

The target question is not whether the model can recover the latent path perfectly with only 30 time points. Rather, the target question is whether reducing T creates regression bias, unstable MCMC behavior, or severe degradation in latent-state recovery. This distinction matters because short time series naturally make path-shape correlation metrics noisier, even when the posterior mean path is close in RMSE.

3 Data calibration

Table 1: Data-scale and covariate diagnostics for Scenario 4D.

Reps	T	Cells	Mean count	Zero prop.	Count CV	x2 mode	Mean lag-1 x2	Mean within-SD x2
10	30	270	202.167	0.48%	1.062	continuous_time	0.423	0.966

The simulated data retain a high-count setting despite the shortened time horizon. The average count is about 202, the zero proportion is below 1%, and the count CV is about 1.06. The fitted data also show the intended time-varying covariate structure: x_2 has positive within-region temporal variability and a finite average lag-1 correlation.

4 Fitting strategy

The model is fitted with γ fixed at 0.8. This keeps the discount-factor learning issue separate from the short-time-series stress test. The fitted parameters include β_0, β_1, β_2, the spatial random effects, τ_ϕ, r, the latent gamma mixing terms κ, and the latent dynamic risk paths λ.

Fit status is classified using guard counts and extreme posterior summaries. A stable replicate has no beta guard, kappa guard, or lambda guard activity and no severe posterior pathology.

Table 2: Fit-status classification across the 10 Scenario 4D replicates.

Status	Replicates	Proportion
Stable	10	100.0%
Soft warning	0	0.0%
Numerical instability	0	0.0%

All 10 replicates are stable. There are no soft warnings and no numerical instabilities. This is the strongest evidence that short $T = 30$ alone does not produce MCMC instability in this design.

5 Replicate-level summaries

The next two tables show replicate-level posterior summaries. They are split into regression/dispersion and latent-state summaries to avoid wide-table overflow in the PDF.

Table 3: Replicate-level regression and dispersion posterior means.

Rep	beta0	beta1	beta2	r	x2 lag-1
1	-1.647	0.482	-0.429	18.087	0.304
2	-1.701	0.502	-0.415	14.903	0.410
3	-1.329	0.513	-0.404	13.316	0.446
4	-1.653	0.509	-0.431	15.027	0.474
5	-1.684	0.512	-0.395	18.582	0.440
6	-1.706	0.525	-0.399	13.992	0.402
7	-2.104	0.531	-0.396	15.442	0.422
8	-1.895	0.539	-0.380	16.028	0.420
9	-1.886	0.459	-0.423	14.506	0.484
10	-1.500	0.493	-0.382	15.468	0.424

Table 4: Replicate-level latent-risk and kappa recovery summaries.

Rep	log-lambda RMSE	cor(log-lambda)	cor(delta log-lambda)	kappa CV	log-kappa RMSE	cor(log-kappa)
1	0.054	0.240	0.035	0.212	0.130	0.863
2	0.050	0.476	0.060	0.232	0.117	0.901
3	0.046	0.349	0.127	0.266	0.093	0.944
4	0.045	0.378	0.199	0.238	0.123	0.897
5	0.039	0.516	0.080	0.221	0.101	0.913
6	0.035	0.755	0.315	0.248	0.107	0.921
7	0.099	0.969	0.510	0.221	0.154	0.844
8	0.067	0.725	0.456	0.221	0.135	0.843
9	0.062	0.577	0.428	0.248	0.149	0.851
10	0.039	0.336	0.100	0.230	0.103	0.913

The regression summaries are stable across replicates. The posterior mean of β_2 ranges from about -0.431 to -0.380, with a cross-replicate standard deviation of about 0.018. This is small relative to the true value -0.4. The latent-path correlation metrics vary more across replicates, which is expected under a short time horizon.

6 Posterior performance by subset

Because all replicates are stable, the all-sampled, stable-only, and stable-plus-soft-warning subsets are identical. The numerical-instability subset is empty.

Table 5: Posterior performance by fit-status subset: data scale and regression recovery.

Subset	n	Mean count	Zero prop.	beta1	beta2	SD beta2	r
All sampled	10	202.167	0.48%	0.506	-0.405	0.018	15.535
Stable	10	202.167	0.48%	0.506	-0.405	0.018	15.535
Stable+soft	10	202.167	0.48%	0.506	-0.405	0.018	15.535
Numerical instability	0	NA	NA	NA	NA	NA	NA

Table 6: Posterior performance by fit-status subset: latent-risk and kappa recovery.

Subset	log-lambda RMSE	cor(log-lambda)	cor(delta log-lambda)	kappa CV	log-kappa RMSE	cor(log-kappa)	Lambda
All sampled	0.053	0.532	0.231	0.234	0.121	0.889	0
Stable	0.053	0.532	0.231	0.234	0.121	0.889	0
Stable+soft	0.053	0.532	0.231	0.234	0.121	0.889	0
Numerical instability	NA	NA	NA	NA	NA	NA	NA

The all-sampled subset has $\bar{\beta}_1 = 0.506$, $\bar{\beta}_2 = -0.405$, and $\bar{r} = 15.535$. These are close to the true values 0.5, -0.4, and 15. The average log-λ RMSE is 0.053, indicating that the scale of the latent dynamic path is recovered well. The average correlation of $\Delta \log \lambda$ is lower, around 0.231, which indicates that fine-scale temporal increments are more difficult to recover with only 30 time points.

7 Regression recovery

Table 7: Regression recovery for the all-sampled subset.

Parameter	Truth	Posterior mean	Across-rep SD	Mean bias	Bias RMSE	Coverage
beta0	-1.712	-1.711	0.215	0.001	0.018	100.0%
beta1	0.500	0.506	0.024	0.006	0.024	100.0%
beta2	-0.400	-0.405	0.018	-0.005	0.018	100.0%

Regression recovery is strong for all three coefficients. The mean bias is essentially zero for β_0, about 0.006 for β_1, and about -0.005 for β_2. Coverage is 100% for all three coefficients across the 10 replicates.

The most important feature is that β_2 is not fragile in this formal S4D setting. Its posterior mean is close to the truth and varies little across replicates. Thus, when the dynamic covariate design is retained and only T is shortened, the model continues to recover both regression effects well.

8 Dispersion and kappa recovery

The posterior mean of r averages 15.535, close to the truth $r = 15$. This indicates that the short time horizon does not produce a strong dispersion bias in this design.

Kappa recovery is also stable. The average posterior kappa CV is about 0.234, the average log-kappa RMSE is about 0.121, and the average correlation of log-kappa is about 0.889. These summaries suggest that the model continues to separate overdispersion from the main latent-risk path reasonably well.

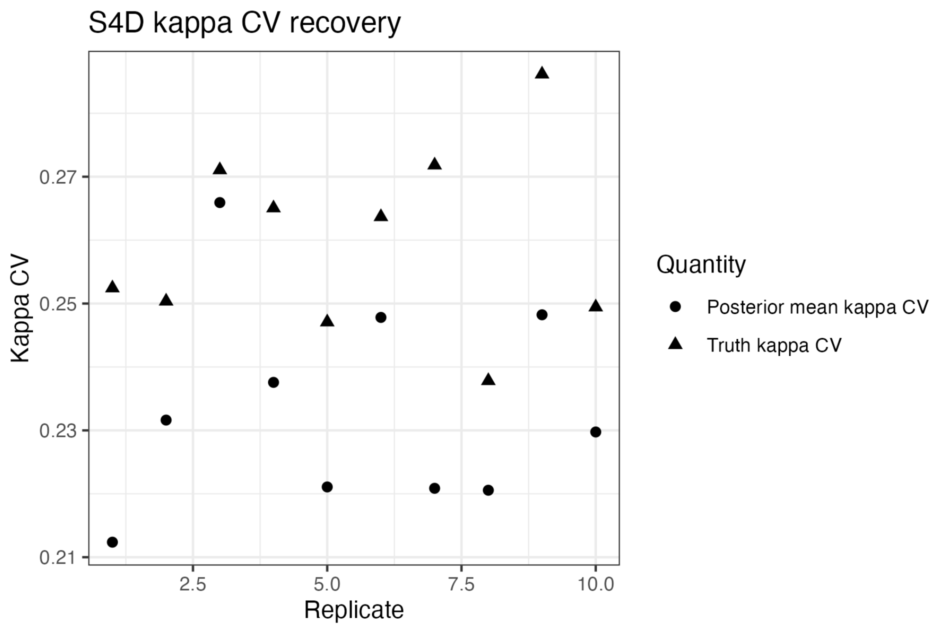

Figure 1: Replicate-level kappa CV posterior summaries.

9 Latent-risk path recovery

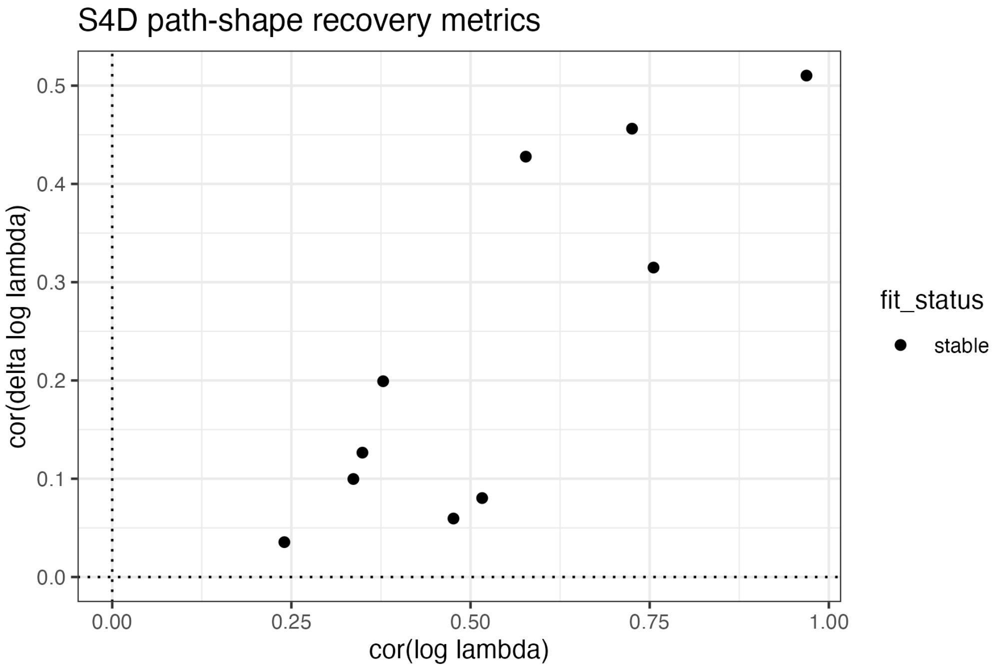

Figure 2: Latent dynamic path recovery by replicate.

The latent dynamic path is recovered well in RMSE but less strongly in correlation-based shape metrics. This pattern is coherent with the design: with $T = 30$, each replicate has fewer time points for learning the path shape, and increments in $\log \lambda$ are especially sensitive to local noise. The short time horizon therefore weakens fine-scale dynamic-shape recovery more than it weakens regression recovery or overall path scale recovery.

10 Why oracle diagnostics are not needed

Oracle diagnostics are most useful when a scenario shows severe bias, instability, or a clear ridge among β, ϕ, λ, and κ. Scenario 4D does not show that pattern. All fits are stable, all guard counts are zero, regression recovery is accurate, and kappa recovery is reasonable. The main limitation is weaker dynamic-shape correlation, which is expected under short T and does not suggest a pathological posterior ridge.

11 Practical implications

The S4D results suggest that the dynamic MSSTNB model is robust to a substantial reduction in temporal length from $T = 100$ to $T = 30$ when counts remain reasonably informative. Shortening T does not produce numerical instability and does not materially harm regression recovery in this design. The main cost is reduced information for detailed latent-path shape recovery, especially for first differences of $\log \lambda$.

For applied use, this implies that short time series can still support stable inference for regression effects and dispersion when the mean count level is adequate. However, one should be more cautious when interpreting high-frequency changes in the latent dynamic risk path from short time series.

12 Final conclusion

Scenario 4D confirms that short temporal length alone is not a major source of posterior failure in the current MSSTNB simulation design. With $T = 30$, all 10 replicates are stable; β_1, β_2, and r are recovered accurately; and kappa recovery remains strong. The principal effect of shortening the time series is a weaker and more variable correlation-based recovery of the latent dynamic path, especially for temporal increments.

13 Suggested reporting language

The following concise statement can be used in the simulation summary:

> In Scenario 4D, we reduced the temporal horizon to $T = 30$ while retaining the dynamic covariate and latent-risk structure. Across 10 replicates, all fits were stable and no numerical guard was triggered. Regression recovery remained accurate, with average posterior means 0.506 for β_1 and -0.405 for β_2, compared with true values 0.5 and -0.4. The average log-λ RMSE was low at 0.053, although correlation-based latent-path summaries were weaker, especially for increments in $\log \lambda$. Thus, short temporal length mainly limits fine-scale dynamic-shape recovery rather than regression or dispersion recovery.